

PROBABILITY AND STATISTICS

Lectures 7-8

Iryna Rozora

DSci in Math, Professor

irrozora@kse.org.ua, rozora.iryana@gmail.com

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Outline I

- 1 Distributions Related to the Normal Distribution
- 2 Student's t test
- 3 Goodness of Fit test (Kolmogorov test)
- 4 Chi squared test
- 5 Problems

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The Normal Distribution

A random variable $X \sim N(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right),$$

where μ = mean, σ^2 = variance. Special case $Z \sim N(0, 1)$

Standardization: If $X \sim N(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Suppose $X_1, \dots, X_n$ are independent and $X_i \sim N(\mu_i, \sigma_i^2)$. Then

$$Y = \sum_{i=1}^n a_i X_i \sim N\left(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2\right).$$

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

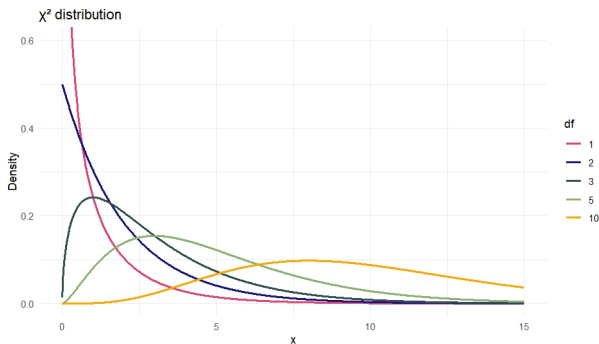
Chi-Square Distribution

If $Z_1, \dots, Z_n$ are independent with $N(0, 1)$, then

$$\chi^2 = \sum_{i=1}^n Z_i^2$$

has a chi-square distribution with n degrees of freedom.

Notation: $\chi^2 \sim \chi_n^2$



Sample Variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Suppose $X_i \sim N(\mu, \sigma^2)$ independently. Then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

Moreover, for a normal sample,

$$\bar{X} \quad \text{and} \quad S^2$$

are independent.

This is a unique property of the normal distribution.

The fact is essential for deriving Student's t -distribution.

Student's t -Distribution

Suppose

$$Z \sim N(0, 1) \quad \text{and} \quad U \sim \chi_n^2$$

are independent.

Then

$$T = \frac{Z}{\sqrt{U/n}} \sim t_n$$

has t -distribution with n df.

Derivation from a Normal Sample

For $X_i \sim N(\mu, \sigma^2)$ we have

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

and

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Using independence,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

F Distribution

Suppose

$$U \sim \chi_{d_1}^2 \quad \text{and} \quad V \sim \chi_{d_2}^2$$

are independent.

Then

$$F = \frac{U/d_1}{V/d_2} \sim F(d_1, d_2)$$

Properties:

- Positive support
- Right-skewed
- Depends on two degrees of freedom
- Used to compare variances
- Used in ANOVA

Ratio of Two Sample Variances

Suppose $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ are independent with $X_i \sim N(\mu_1, \sigma_1^2)$ and $Y_i \sim N(\mu_2, \sigma_2^2)$. Then

$$\frac{(n-1)S_X^2}{\sigma_1^2} \sim \chi_{n-1}^2 \quad \text{and} \quad \frac{(m-1)S_Y^2}{\sigma_2^2} \sim \chi_{m-1}^2.$$

If $\sigma_1^2 = \sigma_2^2$, then

$$\frac{S_X^2}{S_Y^2} \sim F(n-1, m-1)$$

This theorem is used for:

- Equality of variances tests
- ANOVA
- Regression analysis

Key Theorems

1

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

2

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

3

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

4

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

5

$$\frac{S_1^2}{S_2^2} \sim F(n-1, m-1)$$

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Historical Background of the t-Test

Origin of the t-Test

The t-test was developed by **William Sealy Gosset** (1876–1937), a chemist and statistician working for the *Guinness Brewery* in Dublin, Ireland.

- At the beginning of the 20th century, industrial experiments often involved **small sample sizes**.
- Existing methods based on the normal distribution required the population variance to be known, which was rarely the case in practice.
- Gosset derived the distribution of

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}},$$

when the variance is estimated from the sample.

- Because Guinness prohibited employees from publishing research under their own names, Gosset published his work in 1908 under the pseudonym **“Student”**.

One-Sample t-Test

Hypotheses:

$$H_0 : \mu = \mu_0$$

vs

$$H_A : \mu \neq \mu_0 \quad (\mu > \mu_0 \text{ (greater)} \quad \text{or} \quad \mu < \mu_0 \text{ (less)})$$

Test statistic:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

Under H_0 :

$$T \sim t_{n-1}.$$

Reject H_0 if

$$|T| > t_{1-\alpha/2, n-1} \quad (T > t_{1-\alpha, n-1} \quad \text{or} \quad T < -t_{1-\alpha, n-1})$$

One-Sample Test: Actual Computation

$$H_0 : \mu = 4000$$

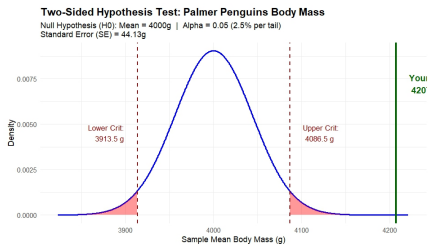
$$H_A : \mu \neq 4000.$$

```
t <- t.test(data$body_mass_g,  
            mu = 4000)  
t  
t$statistic  
t$p.value  
t$conf.int  
t$estimate
```

```
> t
```

One Sample t-test

```
data: data$body_mass_g  
t = 4.6925, df = 332, p-value = 3.952e-06  
alternative hypothesis: true mean is not equal to 4000  
95 percent confidence interval:  
 4120.256 4293.858  
sample estimates:  
mean of x  
 4207.057
```



Since $p\text{-value} < 0.05$, we reject H_0 . There is statistically significant evidence that the mean penguin body mass differs from 4000 g.

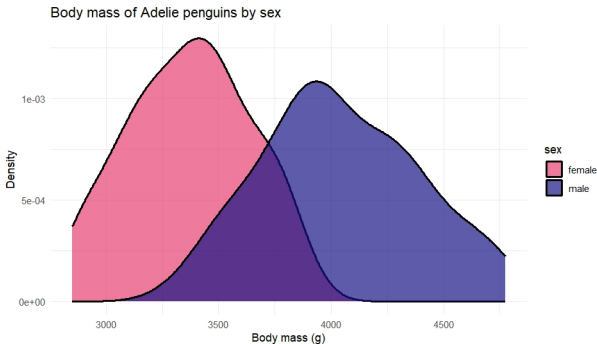
Question:

Do male and female penguins have equal mean body mass? Consider only Adelie penguins.

$$H_0 : \mu_{male} = \mu_{female}$$

vs

$$H_a : \mu_{male} > \mu_{female}$$



Pooled Variance

Assume $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$ are independent with

$$X_i \sim N(\mu_X, \sigma^2), \quad Y_j \sim N(\mu_Y, \sigma^2),$$

A weighted average gives

$$S_p^2 = \frac{(n_1 - 1)S_X^2 + (n_2 - 1)S_Y^2}{n_1 + n_2 - 2}.$$

Statistic:

$$T = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1 + n_2 - 2} \quad \text{under } H_0.$$

Welch Two-Sample Test

No equal variance assumption.

Statistic:

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{S_X^2/n_1 + S_Y^2/n_2}}.$$

Degrees of freedom:

$$\nu = \frac{(S_X^2/n_1 + S_Y^2/n_2)^2}{\frac{(S_X^2/n_1)^2}{n_1-1} + \frac{(S_Y^2/n_2)^2}{n_2-1}}.$$

(Satterthwaite approximation)

Two samples Student T-test in Practice

```
1 male_mass <- data$body_mass_g[
2   data$sex == "male" &
3   data$species == "Adelie"]
4
5 female_mass <- data$body_mass_g[
6   data$sex == "female" &
7   data$species == "Adelie"]
8
9 t.test(
10  male_mass,
11  female_mass,
12  alternative = "greater",
13  var.equal = TRUE)
```

Output:

Two Sample t-test

data: *male_mass* and *female_mass*

$t = 13.126$, $df = 144$, $p\text{-value} < 2.2e-16$

alternative hypothesis: true difference in
means is greater than 0

95 percent confidence interval:

589.5689 Inf

sample estimates:

mean of x mean of y

4043.493 3368.836

Suppose output reports

$$t = 13.126, \quad p < 0.001.$$

Then:

- Strong evidence against H_0 .
- Male body mass is greater than and female body mass.
- The result is statistically significant.
- Confidence interval estimates the magnitude of the difference.

Comparison of t-Tests

Test	Samples	Parameter	DF
One-sample	1	μ	$n - 1$
Two-sample	2 independent	$\mu_1 - \mu_2$	$n_1 + n_2 - 2$
Welch	2 independent	$\mu_1 - \mu_2$	Satterthwaite
Paired	matched pairs	μ_D	$n - 1$

Robustness of the t-Test

The t-test is often robust to moderate violations of normality.

Reasons:

- By the Central Limit Theorem,

$$\bar{X}$$

approaches normality as n increases.

- Symmetric distributions rarely cause problems.
- Moderate skewness is usually acceptable for large samples.

Potential difficulties:

- Strong skewness.
- Heavy tails.
- Extreme outliers.
- Very small samples.

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Kolmogorov test

Task:

Test whether a sample follows a known continuous distribution $F_0(x)$?

$$H_0 : F(x) = F_0(x).$$

$$H_1 : F(x) \neq F_0(x).$$

Statistic:

$$D_n = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)|,$$

where $\hat{F}_n(x)$ is empirical cdf (ecdf)

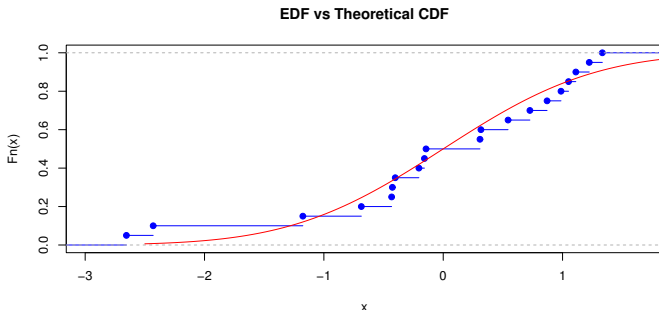
$$\hat{F}_n(x) = \frac{1}{n} \sum_{k=1}^n I_{\{\xi_k \leq x\}}.$$

ECDF

Glivenko-Cantelli theorem: $D_n \rightarrow 0$ as $n \rightarrow \infty$.

Kolmogorov's theorem: $\sqrt{n} \cdot D_n$ tends by distribution to Kolmogorov distribution.

```
x<-rnorm(20, 0, 1)
plot(ecdf(x), col = "blue", main = "ECDF vs Theoretical CDF")
grid <- seq(-2.5, 2.5, length.out = 100)
lines(grid, pnorm(grid, 0, 1), col = "red")
```



Example

In R, the main function for this is `ks.test()`.

Let's consider an example where we test whether a sample belongs to the normal distribution $N(\mu, \sigma^2)$, but with a small “contamination”, to see how the criterion works.

Example

```
# Перевірка на нормальність з "забрудненням"
set.seed(123)
n <- 100
data <- rt(n, df = 2) # t-розподіл (важкі хвости)

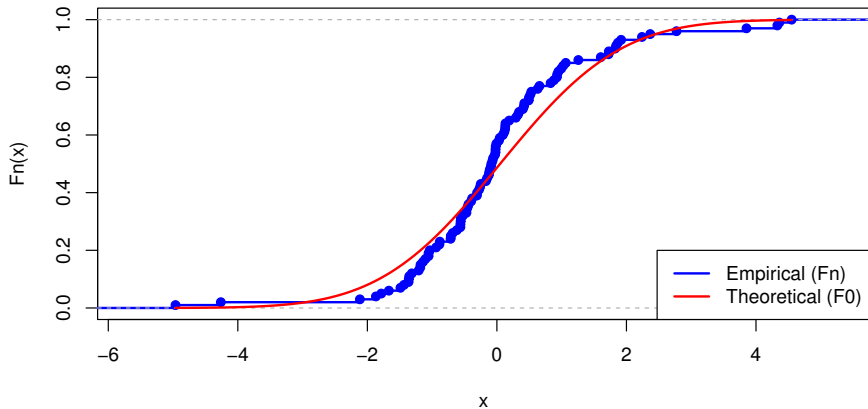
# Виконання KS-тесту
ks_result <- ks.test(data, "pnorm", mean = mean(data), sd = sd(data))
print(ks_result)

# Візуалізація
plot(ecdf(data), col = "blue", main = "EDF vs Theoretical CDF")
grid <- seq(min(data), max(data), length.out = 100)
lines(grid, pnorm(grid, 0, 2, col = "red"))
```

```
print(ks_result)

      Asymptotic one-sample Kolmogorov-Smirnov test
data:  data
D = 0.14431, p-value = 0.03105
alternative hypothesis: two-sided
```

EDF vs Theoretical CDF



Important Notes for Practitioners

- ❶ Ties: If the data have the same values, the p-value becomes approximate. R will issue a warning.
- ❷ Lilliefors Test: If the parameters μ, σ are estimated from the same sample, use the `nortest` package:
 - `lillie.test(x)`
- ❸ For testing normality, the Shapiro-Wilk test usually has higher power.

Shapiro–Wilk Test

Let $\xi_{(1)} \leq \xi_{(2)} \leq \dots \leq \xi_{(n)}$ denote the ordered sample (order statistics) of a sample of size n .

Hypotheses:

- $H_0 : X \sim N(\mu, \sigma^2)$
- $H_1 : X \not\sim N(\mu, \sigma^2)$

Test statistic:

$$W = \frac{(\sum_{i=1}^n a_i \xi_{(i)})^2}{\sum_{i=1}^n (\xi_i - \bar{\xi})^2}$$

where a_i are constants derived from the expected values of the order statistics of a standard normal distribution and are available in tabulated form.

Properties of the Test

- Range of values: $0 < W \leq 1$. Values of W close to 1 provide evidence in favor of H_0 .
- The Shapiro–Wilk test is widely regarded as one of the most powerful tests for assessing normality, often outperforming the Kolmogorov–Smirnov and Anderson–Darling tests for a broad range of alternatives.
- **Limitations:**
 - Highly sensitive for large sample sizes: even minor departures from normality may lead to rejection of H_0 .
 - In many software implementations, the recommended sample size range is $3 \leq n \leq 5000$.

R Implementation: Normally Distributed Data

```
# Generate a sample from a normal distribution
set.seed(456)
x_norm <- rnorm(100, mean = 50, sd = 10)

# Perform the Shapiro-Wilk test
shapiro_result <- shapiro.test(x_norm)

# Display the results
print(shapiro_result)
```

Interpretation: If $p > \alpha$, there is insufficient evidence to reject the assumption of normality.

```
Shapiro-Wilk normality test

data:  x_norm
W = 0.99075, p-value = 0.7253
```


R Implementation: Exponential Distribution

```
# Generate clearly non-normal data
x_exp <- rexp(100, rate = 0.2)

# Perform the test
shapiro_test_exp <- shapiro.test(x_exp)

print(shapiro_test_exp)

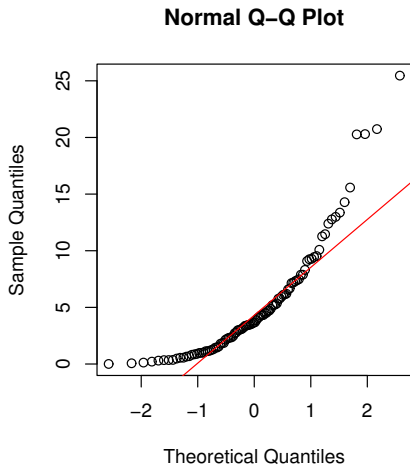
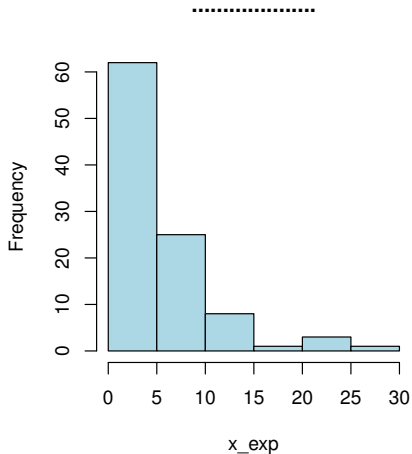
# Visual diagnostics
par(mfrow = c(1, 2))
hist(x_exp, main = "Histogram", col = "lightblue")
qqnorm(x_exp)
qqline(x_exp, col = "red")
```

Shapiro-Wilk normality test

```
data:  x_exp
W = 0.82889, p-value = 2.12e-09
```

Conclusion: Since $p \ll 0.05$, the null hypothesis of normality is rejected.

Example of a Non-Normal Sample



Q-Q Plot as a Complementary Diagnostic Tool

Visual inspection can be used as an informal assessment of normality. A QQ-plot compares the quantiles of a sample to theoretical quantiles of a reference distribution. Points lying on the diagonal indicate a good fit.

- **Q-Q Plot:** If the points lie approximately on a straight line, the data are consistent with a normal distribution.
- The **Shapiro-Wilk test** provides a formal statistical assessment through its p -value.
- For very large samples ($n > 5000$), the `shapiro.test()` function is not applicable. In such cases, the **Anderson-Darling test** (e.g., `ad.test()` from the `nortest` package) is commonly used.

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Chi squared test for Goodness of Fit

The chi-square (χ^2) test is a nonparametric statistical test used to compare observed and expected frequencies.

Common applications:

- Goodness-of-fit test
- Test of independence in contingency tables
- Test of homogeneity

Hypotheses (Goodness-of-Fit):

- H_0 : The data follow the specified distribution.
- H_1 : The data do not follow the specified distribution.

Test Statistic

Suppose the data are grouped into k categories.

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

where

- O_i – observed frequency in category i ;
- E_i – expected frequency under H_0 .

Under H_0 , the test statistic approximately follows a chi-square distribution with

$$df = k - 1 - m$$

degrees of freedom, where m is the number of estimated parameters.

Properties of the Test

- Applicable to categorical or grouped data.
- The test compares observed frequencies with theoretical expectations.
- Larger values of χ^2 indicate greater disagreement between observed and expected frequencies.
- **Assumptions:**
 - Observations are independent.
 - Expected frequencies should generally be at least 5 in each category.
 - Categories must be mutually exclusive.

Chi-Square Goodness-of-Fit Test for a Distribution Function

To test the agreement of a random sample $X = (\xi_1, \dots, \xi_n)$ with a given cumulative distribution function F of a single observation, the real line is partitioned into k disjoint intervals:

$$\mathbb{R} = \bigcup_{i=1}^k \Delta_i,$$

and the histogram counts are computed:

$$\hat{\nu}_{ni} = \sum_{j=1}^n 1_{\{\xi_j \in \Delta_i\}} = |\{j : \xi_j \in \Delta_i\}|, \quad i = \overline{1, k}.$$

The test statistic is

$$\hat{\chi}_{k-1}^2(n) = \sum_{i=1}^k \frac{(\hat{v}_{ni} - nF(\Delta_i))^2}{nF(\Delta_i)}$$

and asymptotically has a chi-square distribution χ_{k-1}^2 . The theoretical probabilities are equal to the increments of the distribution function F over the intervals:

$$F(\Delta_i) = P(\xi_j \in \Delta_i).$$

Decision

Reject H_0 at significance level α if

$$\hat{\chi}_{k-1}^2(n) > \chi_{crit}^2(1 - \alpha, k - 1),$$

where $\chi_{crit}^2(1 - \alpha, k - 1)$ is the $(1 - \alpha)$ -quantile of the chi-square distribution with $k - 1$ degrees of freedom.

Example

Example: A six-sided die was rolled 60 times, producing the following outcomes:

Face	1	2	3	4	5	6
Observed Frequency (O_i)	13	8	12	10	7	10

Task: At the significance level $\alpha = 0.05$, test whether the die is fair, i.e., whether the outcomes follow a uniform distribution.

Formulation of Hypotheses

- H_0 : The die is fair. The probability of each face is $p_i = 1/6$.
- H_1 : The die is biased. The distribution differs from the uniform distribution.

Expected Frequency (E_i):

Since $n = 60$ rolls and there are $k = 6$ categories,

$$E_i = np_i = 60 \cdot \frac{1}{6} = 10.$$

Thus, the expected frequency is 10 for each face.

Computing the Test Statistic

The chi-square goodness-of-fit statistic is

$$\hat{\chi}^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = \frac{(13 - 10)^2}{10} + \dots + \frac{(10 - 10)^2}{10}.$$

Therefore,

$$\hat{\chi}^2 = 0.9 + 0.4 + 0.4 + 0 + 0.9 + 0 = 2.6.$$

- Degrees of freedom:

$$df = k - 1 = 6 - 1 = 5.$$

- Significance level:

$$\alpha = 0.05.$$

- From the chi-square table:

$$\chi_{crit}^2(0.95; 5) \approx 11.07.$$

Conclusion:

Since

Implementation in R

```
# Observed frequencies
observed <- c(13, 8, 12, 10, 7, 10)

# Expected probabilities under H0
probs <- rep(1/6, 6)

# Chi-square goodness-of-fit test
result <- chisq.test(observed, p = probs)

# Display the results
print(result)
```

Output:

X-squared = 2.6, df = 5, p-value = 0.7615

Since the p-value is much larger than 0.05, we do not reject H_0 .

R Implementation: Biased Die

```
# Observed frequencies
observed <- c(5, 8, 10, 14, 23, 40)

chisq.test(x = observed,
           p = rep(1/6, 6))
```

Output:

```
Chi-squared test for given probabilities

X-squared = 47.12,
df = 5,
p-value = 5.5e-09
```

Conclusion:

Since $p < 0.05$, the null hypothesis is rejected. The observed frequencies differ significantly from those expected for a fair die.

Chi-Square Test of Independence

Consider a random sample consisting of bivariate observations (pairs of measurements):

$$X = ((\xi_j, \eta_j), j = \overline{1, n}).$$

The joint distribution of the pairs is unknown. The null hypothesis states that the two variables in each pair are independent.

By grouping the data, the problem can be reduced to the case where the sample values belong to finite sets:

$$\xi_1 \in \{x_1, \dots, x_k\}, \quad \eta_1 \in \{y_1, \dots, y_m\}.$$

In this setting, the null hypothesis can be written as

$$H_0 : P_{\theta}(\xi_1 = x_i, \eta_1 = y_l) = p_i q_l, \quad i = \overline{1, k}, \quad l = \overline{1, m}.$$

the joint frequencies

$$\nu_{il} = \sum_{j=1}^n 1_{\{\xi_j = x_i, \eta_j = y_l\}} = |\{j : \xi_j = x_i, \eta_j = y_l\}|,$$

Contingency Table

The table of frequencies

$$(\nu_{il}, i = \overline{1, k}, l = \overline{1, m})$$

is called a *contingency table*. It summarizes the joint distribution of the categorical variables represented by the rows and columns.

General Contingency Table

	y_1	y_2	$\cdots$	y_m	Total
x_1	ν_{11}	ν_{12}	$\cdots$	ν_{1m}	$\nu_{1\cdot}$
x_2	ν_{21}	ν_{22}	$\cdots$	ν_{2m}	$\nu_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
x_k	ν_{k1}	ν_{k2}	$\cdots$	ν_{km}	$\nu_{k\cdot}$
Total	$\nu_{\cdot 1}$	$\nu_{\cdot 2}$	$\cdots$	$\nu_{\cdot m}$	n

Here,

$$\nu_{il} = |\{j : \xi_j = x_i, \eta_j = y_l\}|$$

is the observed frequency in cell (i, l) ,

$\nu_{i\cdot} = \sum_{l=1}^m \nu_{il}$ is the total frequency in row i , $\nu_{\cdot l} = \sum_{i=1}^k \nu_{il}$ is the total frequency in column l , and $n = \sum_{i=1}^k \sum_{l=1}^m \nu_{il}$ is the sample size.

Therefore, the statistic for testing independence is

$$\hat{\chi}^2(\hat{\theta}_n) = \sum_{i=1}^k \sum_{l=1}^m \frac{(\hat{v}_{il} - \hat{v}_{i\cdot} \hat{v}_{\cdot l} / n)^2}{\hat{v}_{i\cdot} \hat{v}_{\cdot l} / n},$$

where

$$\hat{v}_{i\cdot} = \sum_{l=1}^m \hat{v}_{il}, \quad \hat{v}_{\cdot l} = \sum_{i=1}^k \hat{v}_{il}$$

are the row and column totals.

Under the null hypothesis,

$$\hat{\chi}^2 \xrightarrow{d} \chi_{(k-1)(m-1)}^2, \quad n \rightarrow \infty.$$

Decision

Reject H_0 at significance level α if

$$\hat{\chi}^2 > \chi_{crit}^2(1 - \alpha, (k - 1) \cdot (m - 1)),$$

By the asymptotic theory of the modified chi-square statistic, the significance level of the test is approximately equal to α for sufficiently large sample sizes.

Remark

When $m = k = 2$, the chi-square statistic has only one degree of freedom, and all terms in the sum are proportional. In this case, it is common to apply **Yates' continuity correction**, replacing the statistic by

$$\hat{\chi}^2(\hat{\theta}_n) = \frac{(\hat{\nu}_{11} - \hat{\nu}_{1.}\hat{\nu}_{.1}/n - \frac{1}{2})^2}{\hat{\nu}_{1.}\hat{\nu}_{.1}\hat{\nu}_{2.}\hat{\nu}_{.2}/n^3}.$$

Example

A study investigates whether gender is associated with preference for a sport (football or tennis).

Observed Frequencies (O_{ij}):

	Football	Tennis	Total
Men	20	10	30
Women	10	20	30
Total	30	30	60

Null Hypothesis (H_0): Gender and sport preference are independent.

Alternative Hypothesis (H_1): Gender and sport preference are associated.

Computation of the Test Statistic

$$\begin{aligned}\hat{\chi}^2(\hat{\theta}_n) &= \sum_{i=1}^k \sum_{l=1}^m \frac{(\hat{v}_{il} - \hat{v}_{i.}\hat{v}_{.l}/n)^2}{\hat{v}_{i.}\hat{v}_{.l}/n} \\ &= 2 \cdot \frac{(20 - 30 \cdot 30/60)^2}{30 \cdot 30/60} + 2 \cdot \frac{(10 - 30 \cdot 30/60)^2}{30 \cdot 30/60} = 6.667.\end{aligned}$$

- Degrees of freedom: $df = (k - 1)(m - 1) = (2 - 1)(2 - 1) = 1$.
- Significance level: $\alpha = 0.05$.
- Critical value: $\chi_{crit}^2(0.95; 1) = 3.84$.

Conclusion:

Since $\hat{\chi}^2 = 6.667 > 3.84 = \chi_{crit}^2$, we reject the null hypothesis H_0 .

There is statistically significant evidence of an association between gender and sport preference.

Implementation in R

```
# Create the contingency table
data_matrix <- matrix(c(20, 10, 10, 20),
                      nrow = 2,
                      byrow = TRUE)

colnames(data_matrix) <- c("Football", "Tennis")
rownames(data_matrix) <- c("Men", "Women")

# Chi-square test of independence
# correct = FALSE disables Yates' correction
# to match the manual calculations
result <- chisq.test(data_matrix, correct = FALSE)

# Display the test results
print(result)

      Pearson's  $\chi^2$  Chi-squared test

data: data_matrix
 $\chi^2$  = 6.6667, df = 1, p-value = 0.009823
```


Outline

- 1 Distributions Related to the Normal Distribution
- 2 Student's t test
- 3 Goodness of Fit test (Kolmogorov test)
- 4 Chi squared test
- 5 Problems**

Problem 1. Shapiro–Wilk Test (Iris dataset)

Problem 1

The Iris dataset contains measurements of 150 flowers from three species. We will investigate whether the sepal length of Setosa flowers is normally distributed.

- Consider only Setosa flowers.
- Formulate the hypotheses.
- Draw a histogram.
- Draw a Q-Q plot.
- Perform the Shapiro–Wilk test.
- Interpret the p-value at significance level $\alpha=0.05$. Is the normality assumption reasonable?

Problem 2. Kolmogorov–Smirnov Test

The Palmer Penguins dataset contains measurements of penguins from Antarctica.

We will investigate whether the body mass of Adelie penguins follows a normal distribution.

Problem 2. Kolmogorov–Smirnov Test (Palmer Penguins)

- Use only Adelie penguins.
- Remove missing values.
- Standardize body mass.
- Test

$$H_0 : X \sim N(0, 1)$$

using the Kolmogorov–Smirnov test.

Interpret the result.

- Compare your conclusion with the Shapiro–Wilk test.

Problem 3. Chi-Square Goodness-of-Fit Test

The Diamonds dataset contains information about almost 54,000 diamonds.

We will investigate whether diamond cuts are equally common.

Cut Categories: Fair; Good; Very Good; Premium; Ideal

Problem 3. Chi-Square Goodness-of-Fit Test (Diamonds dataset)

Suppose all five categories should occur equally often. Test this claim.

- State hypotheses.
- Compute observed frequencies.
- Compute expected frequencies.
- Perform a Chi-square goodness-of-fit test.
- Interpret the result.

Problem 4. Chi-Square Test, Independence

Problem 4.

Use the built-in MASS::survey dataset. The dataset contains information collected from university students.

We want to investigate whether smoking habits are associated with gender.

- Create a contingency table(Gender (Sex), Smoking habit (Smoke))
- State the hypotheses
- Perform the Chi-square test
- Make a decision at $\alpha = 0.05$
- Examine expected frequencies
- Visualize the data
- Which smoking category contributes the most to the Chi-square statistic?

Questions?